Summary in Graph

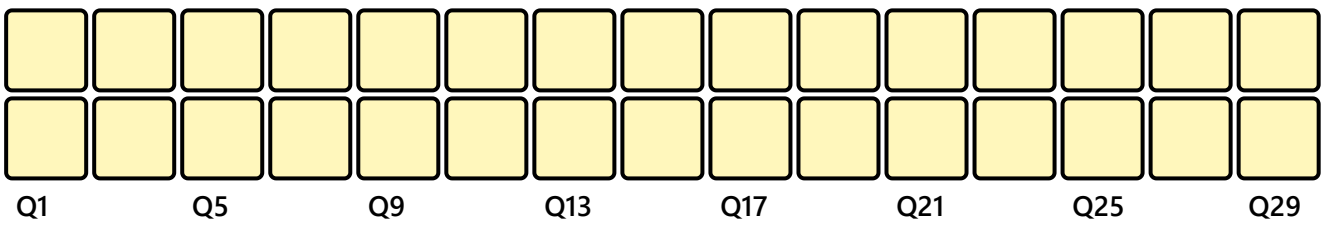
Exam Summary (GATE Overflow CS Test Series 2026 | Algorithms | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (6) out of 59 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Algorithms

If algorithm X is having better asymptotic time complexity than algorithm Y, then which of the following is/are TRUE? (Mark all choices which are CORRECT)

A. X will outperform Y for all inputs
B. X will outperform Y for all small inputs
C. X will outperform Y for all large inputs

D. X will outperform Y for all inputs above 1 million in size

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Suppose you are given the following set of keys to insert into a hash table that holds exactly 10 values: $13, 107, 49, 50, 64, 98, 16, 33$. Which of the following best demonstrates the contents of the hash table after all the keys have been inserted using linear probing and $\mod 10$ as hash function

A. $50 \; - \; - \; 13 \; 64 \; 33 \; 16 \; 107 \; 98 \; 49$
B. $50 \; 33 \; - \; 13 \; 64 \; - \; 16 \; 107 \; 98 \; 49$
C. $50 \; - \; 33 \; 13 \; 64 \; - \; 16 \; 107 \; 98 \; 49$
D. $50 \; - \; - \; 33 \; 64 \; 13 \; 16 \; 107 \; 98 \; 49$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

An array of 8 integers is being sorted by the heapsort algorithm. After the initial phase of the algorithm (constructing the heap), which of the following is a possible ordering for the array? (Mark all the appropriate choices)

A. $2 \; 6 \; 11 \; 13 \; 21 \; 41 \; 43 \; 92$
B. $2 \; 13 \; 6 \; 43 \; 92 \; 11 \; 21 \; 41$
C. $2 \; 6 \; 11 \; 41 \; 92 \; 13 \; 21 \; 43$
D. $2 \; 6 \; 11 \; 21 \; 11 \; 41 \; 43 \; 13$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following is/are TRUE about Huffman Coding? (Mark all the appropriate choices)

A. Huffman coding may become lossy in some cases
B. Huffman Coding does not always have an exact solution
C. In Huffman coding, no code is prefix of any other code
D. There exists a greedy algorithm to do Huffman coding

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the following functions

$$\log_{10} n, \log_2 n, \lceil \log_2 n \rceil, \lfloor \log_2 n \rfloor$$

Which of the following is/are TRUE? (Mark all the appropriate choices)

A. $\log_{10} n = O(\log_2 n)$
B. $\log_2 n = \Omega\lfloor \log_2 n \rfloor$
C. $\lceil \log_2 n \rceil = O(\lfloor \log_2 n \rfloor)$
D. $\lceil \log_2 n \rceil = \Omega(\lfloor \log_2 n \rfloor)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

Consider a min heap implemented as an array with the elements $1, 2, 3, 6, 7, 8, 9, 10$. The level (starting from 1) in which 7 is stored is _____

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

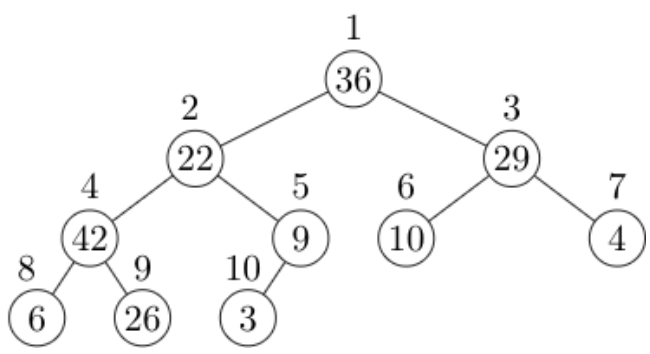
It is required to sort a large number of records of employee details based on the employee IDs. Assuming the records are already sorted on the basis of employee names and it is required to maintain name as the secondary sort order (similar to order by id, name in SQL), which of the following sorting algorithms is/are the most appropriate for this?

A. Merge Sort
B. Selection Sort
C. Quick Sort
D. Heap Sort

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

Consider the binary tree shown below, which is an almost max-heap with the node 22 violating the max-heap property. Once heapify procedure is applied to it, which position will it be in?



Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

In a hashtable with 20 slots 30 records are inserted with collisions being resolved by chaining. What is the expected number of key comparisons in an unsuccessful search assuming uniform hashing?

Your Answer: | Correct Answer: 1.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Which of the following is/are the possible array contents after second pass of Quick Sort for the following initial ordering assuming first element is taken as pivot?

$$34, 8, 64, 51, 32, 21$$

A. $8, 21, 32, 34, 51, 64$
B. $8, 32, 34, 51, 64, 21$
C. $8, 34, 51, 64, 32, 21$
D. $8, 34, 64, 51, 32, 21$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Given an array of n elements, you have to design an algorithm to find the k smallest elements in sorted order. The time complexity of the best such algorithm assuming comparison based sorting will be

A. $O(k \log n)$
B. $\Theta(n + k \log k)$
C. $\Theta(n^2)$
D. $\Theta(n \log k)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose that you store 6 records in a hash table of size 8 by chaining, and suppose that you have a good hash function so that the probability that a key is hashed into any of the 8 slots is $1/8$. For a particular slot in the hash table, what is the probability that this slot is empty, that is, none of the 6 keys hashes into this slot?

A. 0.125^6
B. 0.875^6
C. 0.166^8
D. 0.833^8

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Suppose that you store 5 records in a hash table of size 9 by using open addressing with linear probing, and suppose that you have a good hash function so that the probability that a key is hashed into any of the 9 slots is $1/9$. For a particular slot in the hash table, what is the probability that this slot is empty, that is, none of the 5 keys hashes into this slot (rounded to 2 decimal points)?

Your Answer: | Correct Answer: 0.44:0.45 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following C function and an input array to it as $[1, 4, 10, 12]$ and len as 4. The return value of the function will be

```
int foo(int arr[], int len) {
    int result = 0;
    for (int i = 0; i < len; i++) {
        result += arr[i] * (i + 1) * (len - i);
    }
    return result;
}
```

Your Answer: | Correct Answer: 136 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15

Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

You are given an empty hash table of size 6 that uses open addressing. The following sequence of keys is to be inserted:

$$32\ 7\ 14\ 22\ 29\ 3$$

If linear probing with $h(x) = x\%6$; is used, which all elements will cause a collision? (Mark all choices which are CORRECT)

A. 3
B. 14
C. 29
D. 22

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16

Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following C function:

```
int findme(int n)
{
    if(n <= 0 ) return 1;
    return findme(n/2) + findme(n-1);
}
```

What will be the output if the above function is called with argument 12?

Your Answer: | Correct Answer: 94 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17

Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

For quick sort pivot selection you are provided with an buggy implementation of Median select, which can return any element between $(n/3)^{rd}$ smallest and $(2n/3)^{rd}$ smallest in $O(n)$ time. With this function to choose the pivot, which of the following is/are TRUE for the QuickSort implementation? (Mark all CORRECT choices)

A. In best case it can run in $O(n)$ time
B. In worst case it will take $\Theta(n^2)$ time
C. In worst case it will take $\Theta(n \log n)$ time
D. In best case, it will run in $\Omega(n \log n)$ time

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

What is the time complexity of following code if $foo(n)$ takes $\Theta(\log n)$ time to execute?

```
int i, j;
for (i = n / 2; i <= n; i++) {
    for (j = 1; j <= n; j = j * 2) {
        foo(j);
    }
}
```

A. $\Theta(n)$
B. $\Theta(n \lg n)^2$
C. $\Theta(n(\lg n)^2)$
D. $\Theta(n^2 \lg n)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

The return value, time complexity and space complexity of following code are

```
int foo(int n)
{
    int a = 0;
    while (n > 0) {
        a += n;
        n /= 2;
    }
return a;
}
```

A. $\Theta(\log n), \Theta(\log n), \Theta(\log n)$
B. $\Theta(n), \Theta(\log n), O(1)$
C. $\Theta(\log n), \Theta(\log n), O(1)$
D. $\Theta(n^2), O(n), O(1)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Suppose you are having 5 notes each with denominations of $2000, 500, 300$ and 50 and you are asked to make a sum of Rs. 4650. For each note you choose you are asked to pay a service charge of Rs. 2.5 which is irrespective of the denomination of the note. What will be the minimum amount in rupees you'll have to pay as service charge?

Your Answer: | Correct Answer: 12.5-12.6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE? (Mark all CORRECT choices)

A. The expected number of comparisons in a successful linear search is n
B. The expected number of comparisons in a successful binary search is $O(\log n)$
C. The best case runtime of linear search is asymptotically better than binary search
D. In worst case binary search can lead to $\Omega(n)$ comparisons

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Suppose the letters a, b, c, d have probabilities $1/3, 1/6, 1/5, 3/10$ respectively. What is the average length of Huffman codes (correct to 1 decimal point)?

Your Answer: | Correct Answer: 2.0-2.0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

You are given an empty hash table of size 7 that uses open addressing. The following sequence of keys is to be inserted:

$$13\ 18\ 25\ 11\ 20\ 29$$

If hash function $h_1(x) = x\%7$; and double hashing with $h_2(x) = x/7 + 1$ is followed, which of the inserted elements causes a collision? (Mark all the appropriate choices)

A. 25
B. 11
C. 20
D. 29

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE? (Mark all the appropriate choices)

A. The worst case runtime for searching in a binary tree can be $\Omega(n)$.
B. The minimum number of comparisons to sort 5 elements is 4.
C. There can be at most $\left\lfloor \frac{n}{2^{h-1}} \right\rfloor$ nodes of height h (h starting from 1) in any n element heap.
D. The time to find the maximal element in a min-heap of $2^n - 1$ elements is $\Omega(2^n)$.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Four matrices M_1, M_2, M_3 and M_4 of dimensions $p \times q, q \times r, r \times s$ and $s \times t$ respectively can be multiplied is several ways with different number of total scalar multiplications. For example, when multiplied as $((M_1 \times M_2) \times (M_3 \times M_4))$, the total number of multiplications is $pqr + rst + prt$. When multiplied as $(((M_1 \times M_2) \times M_3) \times M_4)$, the total number of scalar multiplications is $pqr + prs + pst$. If $p = 20, q = 90, r = 10, s = 15$ and $t = 40$, then the minimum number of scalar multiplications needed is

Your Answer: | Correct Answer: 32000 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Maximum Subarray Sum problem is to find the subarray with maximum sum. For example, given an array $\{12, -13, -5, 25, -20, 30, 10\}$, the maximum subarray sum is 45. The best possible algorithm to compute the maximum subarray sum will run in

(Mark all the appropriate choices)

A. $O(n)$
B. `$\Omega(n \log n)$
C. $O(\log n)$
D. $O(n^2)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

You are given an array A of n elements. The best possible algorithm to find two numbers in A which sum to the largest number in A will run in (assuming comparison based sorting)

(Mark all CORRECT choices)

A. $O(n)$
B. `$\Omega(n \log n)$
C. $\Omega(n^2 \log n)$
D. $O(n^2)$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the code given below, which runs insertion sort:

```
void insertionSort(int arr[], int size)
{
    int i, j, value;
    for (i = 1; i < size; i++)
    {
        value = arr[i];
        j = i;
        while (________ )
        {
            arr[j] = arr[j - 1];
            j = j - 1;
        }
        arr[j] = value;
    }
}
```

Which condition will correctly implement the while loop?

A. $(j > 0)$ || $(arr[j-1] > value)$
B. $(j > 0)$ && $(arr[j-1] > value)$
C. $(j > 0)$ && $(arr[j+1] > value)$

D. $(j > 0)$ && $(arr[j+1] < value)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following C function for arguments $m, n > 1$

```
int foo(int n, int m)
{
    while (m != n)
    {
        if(m > n)
            m = m - n;
        else
            n = n - m;
    }
    return n;
}
```

Which of the following is/are true about the above function? (Mark all CORRECT choices)

A. If $m = 9, n = 13$ the function returns 1.
B. The asymptotic time complexity of the function is $O(\log(\max(m, n)))$
C. For any positive integers m, n the function returns the least common multiple of m, n.
D. The space complexity of the function is $\Omega(\log(\min(m, n)))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE? (Mark all CORRECT choices)

A. When 90% of the input is already in sorted order, insertion sort will do less number of comparisons than merge sort for all sufficiently large inputs
B. If the number of inversions in an array is $O(n)$, then insertion sort will be having fewer comparisons than quick sort for all inputs
C. If the number of inversions in an array is $O(n)$, bubble sort will be having fewer number of comparisons than merge sort for all inputs
D. If the number of inversions in an array is $O(n)$, heap sort will be having fewer number of comparisons than insertion sort for all inputs

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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